The No-U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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MCMC & Hamiltonian Monte Carlo

The challenge

- Exact posterior inference is often intractable.
- MCMC is asymptotically unbiased and broadly applicable [Nea93].
- Random-walk & Gibbs [Met+53; GG84] struggle in high-dimensional or correlated spaces.

Hamiltonian Monte Carlo

Auxiliary momentum $r \sim \mathcal{N}(0, I) +$ leapfrog

$$p(\theta, r) \propto \exp\left\{\mathcal{L}(\theta) - \tfrac{1}{2} r^\top r\right\}$$

Volume-preserving means proposals can be **distant** yet **highly probable**.

The HMC Tuning Problem

Step size ε

Too small $\Rightarrow$ wasted computation

Too large $\Rightarrow$ high rejection rate

Leapfrog steps L

Too small $\Rightarrow$ random walk behaviour

Too large $\Rightarrow$ trajectory loops back

Good L requires costly preliminary runs and expert knowledge, a key barrier to adoption.

NUTS addresses both [HG+14]

- **Adaptive L:** No-U-Turn criterion stops when trajectory doubles back.
- **Adaptive ε:** Dual averaging during warm-up [Nes09].

The No-U-Turn Criterion

Key idea: Stop when further steps move the proposal *back* towards the start.

Monitor $\frac{d}{dt}\frac{1}{2}\|\tilde{\theta} - \theta\|^2$: when negative, the trajectory is doubling back.

U-turn condition on leaves (θ^-, r^-), (θ^+, r^+)

$$(\theta^+ - \theta^-) \cdot r^- < 0 \quad \text{or} \quad (\theta^+ - \theta^-) \cdot r^+ < 0$$

Naive rule $\Rightarrow$ breaks time-reversibility. NUTS instead builds trajectories **forwards and backwards** via recursive doubling, augmented with a slice variable:

$$u \sim \mathrm{Uniform}\big([0, \exp\{\mathcal{L}(\theta) - \tfrac{1}{2} r \cdot r\}]\big)$$

Recursive Doubling & Efficient Kernel

Trajectory construction

1. Draw $v_j \sim \mathrm{Uniform}(\{-1, +1\})$
2. Take 2^j leapfrog steps of size $v_j \varepsilon$
3. Grow balanced binary tree of states
4. Halt on U-turn or instability

States from a halting step are excluded from $\mathcal{C}$ to preserve detailed balance.

Memory: $\mathcal{O}(2^j) \to \mathcal{O}(j)$

Partition $\mathcal{C} = \mathcal{C}^{\mathrm{old}} \cup \mathcal{C}^{\mathrm{new}}$ then propose $w' \in \mathcal{C}^{\mathrm{new}}$, accept with:

$$\min\left(1, \frac{|\mathcal{C}^{\mathrm{new}}|}{|\mathcal{C}^{\mathrm{old}}|}\right)$$

Satisfies detailed balance w.r.t. uniform over $\mathcal{C}$.

Adaptive Step Size: Dual Averaging

During warm-up only, drive acceptance towards target δ:

$$x_{t+1} = \mu - \frac{\sqrt{t}}{\gamma(t+t_0)} \sum_{i=1}^{t} H_i, \qquad \bar{x}_{t+1} = t^{-\kappa} x_{t+1} + (1 - t^{-\kappa}) \bar{x}_t$$

- $x = \log \varepsilon, \quad \mu = \log(10\varepsilon_0)$ encourages larger step sizes
- Fixed hyperparameters: $\gamma = 0.05, \;\; t_0 = 10, \;\; \kappa = 0.75$

No-U-Turn criterion + dual averaging $\Rightarrow$ NUTS is **entirely tuning-free**.

The Transition Kernel & Detailed Balance

To reduce memory from $\mathcal{O}(2^j)$ to $\mathcal{O}(j)$, NUTS partitions the candidate set: $\mathcal{C} = \mathcal{C}^{\text{old}} \cup \mathcal{C}^{\text{new}}$.

The Proposal:

- From state $w \in \mathcal{C}^{\text{old}}$, propose a jump to $w' \in \mathcal{C}^{\text{new}}$ uniformly.
- Accept with probability: $\min\left(1, \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right)$

Proof of Detailed Balance (Intuition)

We must satisfy: $p(w|\mathcal{C})T(w'|w,\mathcal{C}) = p(w'|\mathcal{C})T(w|w',\mathcal{C})$

Substituting the prior $p(w|\mathcal{C}) = \frac{1}{|\mathcal{C}|}$ and the acceptance rates:

$$\frac{1}{|\mathcal{C}||\mathcal{C}^{\text{new}}|} \min\left(1, \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right) = \frac{1}{|\mathcal{C}||\mathcal{C}^{\text{old}}|} \min\left(1, \frac{|\mathcal{C}^{\text{old}}|}{|\mathcal{C}^{\text{new}}|}\right)$$

Proving the kernel preserves the target distribution.

Evaluating NUTS: The Paper's Benchmarks

The paper tests NUTS on four targets spanning low to very high dimensions, measuring **minimum ESS per gradient evaluation** across all dimensions.

250-D	25-D	302-D	3001-D
Multivariate Normal	Logistic Regression	Hierarchical LR	Stochastic Volatility

NUTS matches or exceeds best-tuned HMC on **all four benchmarks** without any trajectory-length tuning, up to 2× on the Gaussian and 1.5× on the stochastic volatility model.

But all four targets are **unimodal or mildly structured**. How does NUTS fare on posteriors with genuinely complex geometry?

Beyond the Paper: Three Stress Tests

We probe three failure modes of HMC that are absent from the original evaluation:

Multimodality

BNN, 191-D

$5! \times 5! =$

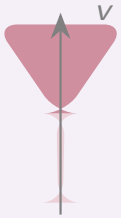
14400 modes

Dense Correlations

GP Classif., 150-D

dense 150×150 cov.

Varying Geometry



Neal's Funnel, 10-D

spatially varying σ

Stress Test 1: Bayesian Neural Network (191-D)

Setup: BNN $(30 \to 5 \to 5 \to 1)$, $\mathcal{N}(0, 100)$ priors on all weights, UCI Breast Cancer dataset. Weight-permutation symmetries create $5! \times 5! = 14{,}400$ equivalent modes.

NUTS explores a **wider region of weight space**; HMC stagnates in a single mode.

NUTS fails to achieve higher ESS per gradient than a well-tuned HMC (0.03×10^{-4} vs. 1.57×10^{-4} for HMC in our tests).

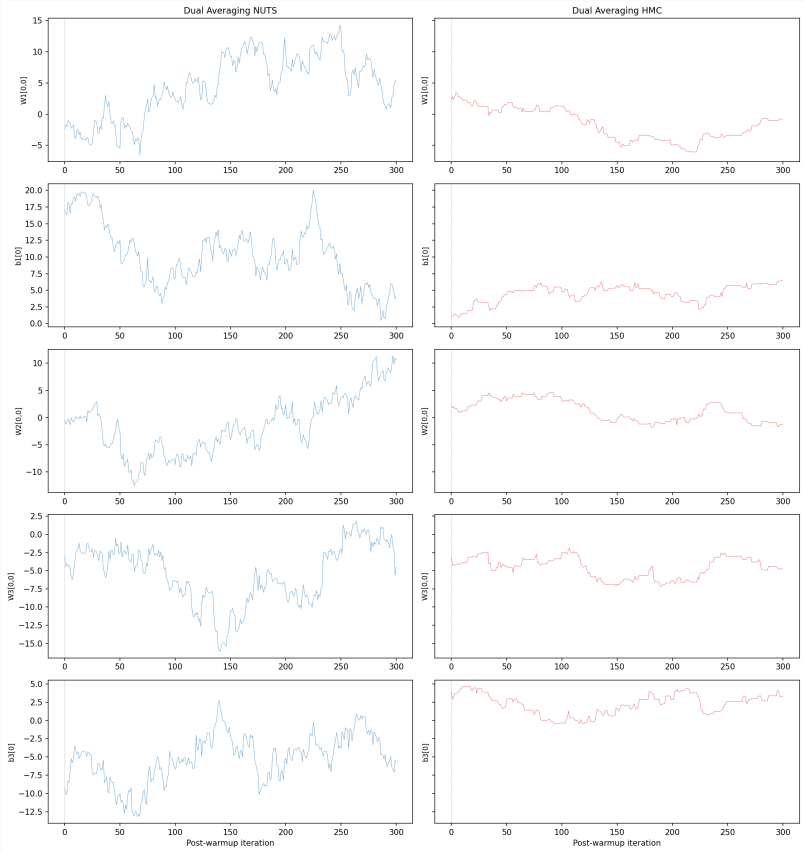


Figure 1: Trace plots: NUTS (blue) covers a broader range of weight values than HMC (red).

Stress Test 2: GP Classification (150-D)

Setup: RBF kernel ($\ell = 1$, $\sigma_f^2 = 1$), 150 California Housing points, fully dense K. The gradient $\nabla \log p(f) = -K^{-1} f + y \odot \sigma(-y \odot f)$ requires a dense solve at every leapfrog step.

Nuanced result

NUTS: 2.14×10^{-3} ESS/grad HMC: 1.52×10^{-3} ESS/grad

Modest 40% gain — but NUTS uses $4\times$ more gradient evaluations. The dense covariance induces long trajectories before a U-turn is detected. The advantage over well-tuned HMC is narrow here.

Stress Test 3: Neal's Funnel (10-D)

$$v \sim \mathcal{N}(0, 9), \qquad x_i \mid v \sim \mathcal{N}(0, e^v), \quad i = 1, \ldots, 9$$

The challenge

$v \to -\infty$: $e^v \to 0$, needs tiny ε

$v > 0$: wide top, needs large ε

HMC's **fixed step size** cannot handle both.

Result

NUTS recovers $\mathrm{Var}(v) \approx 9$ (truth).

HMC is confined and severely underestimates $\mathrm{Var}(v)$.

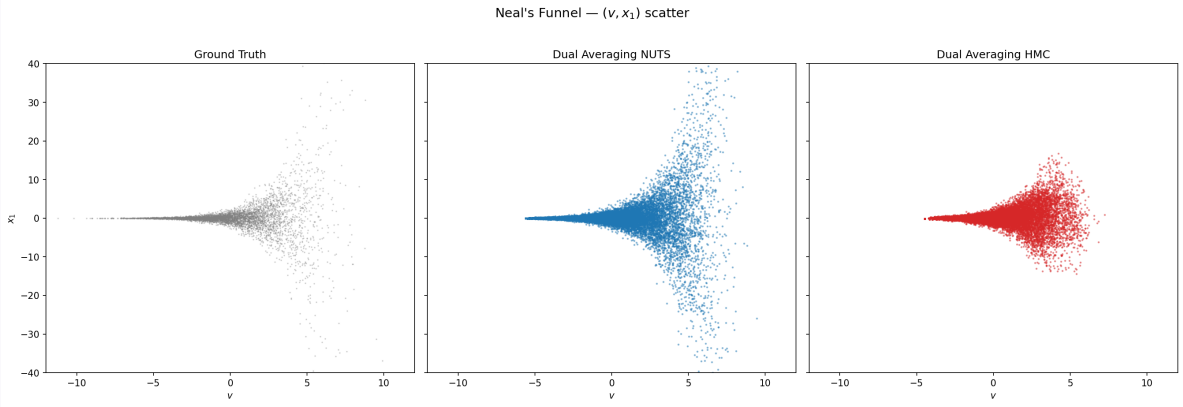


Figure 2: Left: ground truth. **Centre:** NUTS covers the full funnel. **Right:** HMC fails to fully explore the funnel.

Funnel: Trace Plots & Marginal

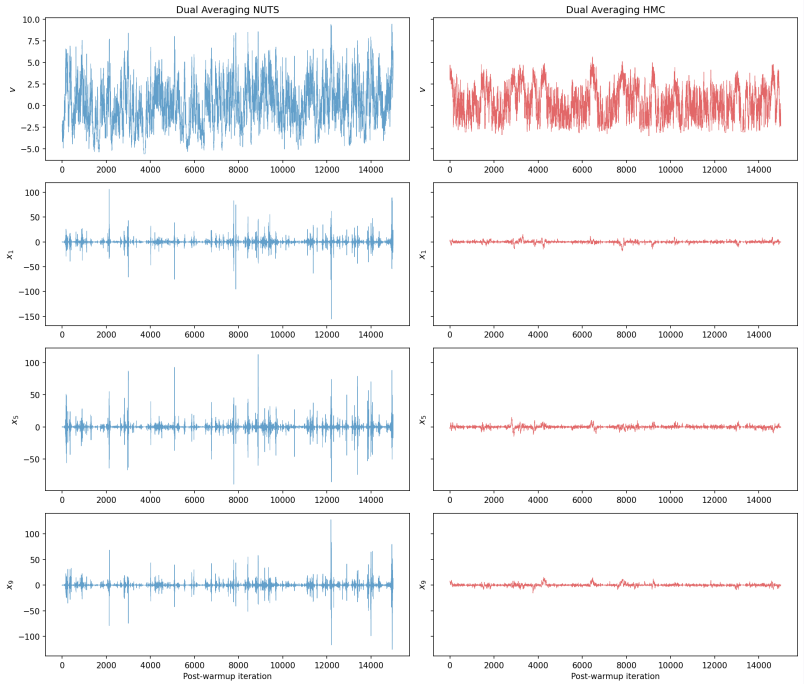


Figure 3: NUTS mixes freely; HMC shows flat segments (rejected proposals).

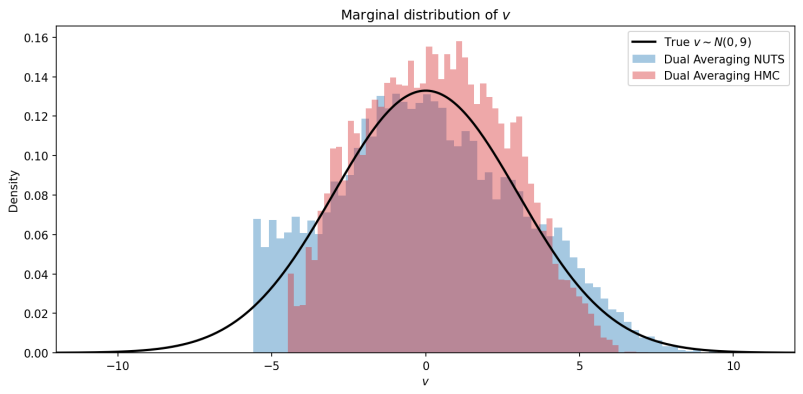


Figure 4: Marginal of v: NUTS recovers the truth; HMC misses $v < 0$ entirely.

Summary & Takeaways

Paper strengths

- Resolves HMC's key practical limitation rigorously
- Recursive doubling preserves detailed balance
- Dual averaging $\Rightarrow$ essentially tuning-free

Limitations

- Dual-averaging hyperparameters empirically justified only
- Guarantees mainly asymptotic
- Original experiments miss pathological geometries

Our contributions

- **Proofs** of uniform $\mathcal{C}$ distribution and detailed balance (omitted in paper)
- **Reproduced** all 4 benchmarks; NUTS matches or exceeds best-tuned HMC

Bibliography i

[Met+53] Nicholas Metropolis et al. **"Equation of state calculations by fast computing machines".** In: *The journal of chemical physics* 21.6 (1953), pp. 1087–1092.

[GG84] Stuart Geman and Donald Geman. **"Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images".** In: *IEEE Transactions on pattern analysis and machine intelligence* 6 (1984), pp. 721–741.

[Nea93] Radford M Neal. **"Probabilistic inference using Markov chain Monte Carlo methods".** In: (1993).

Bibliography ii

[Nes09] Yurii Nesterov. **"Primal-dual subgradient methods for convex problems".** In: *Mathematical programming* 120.1 (2009), pp. 221–259.

[HG+14] Matthew D Hoffman, Andrew Gelman, et al. **"The No-U-Turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo.".** In: *J. Mach. Learn. Res.* 15.1 (2014), pp. 1593–1623.

Appendix: Benchmarks 2 & 3

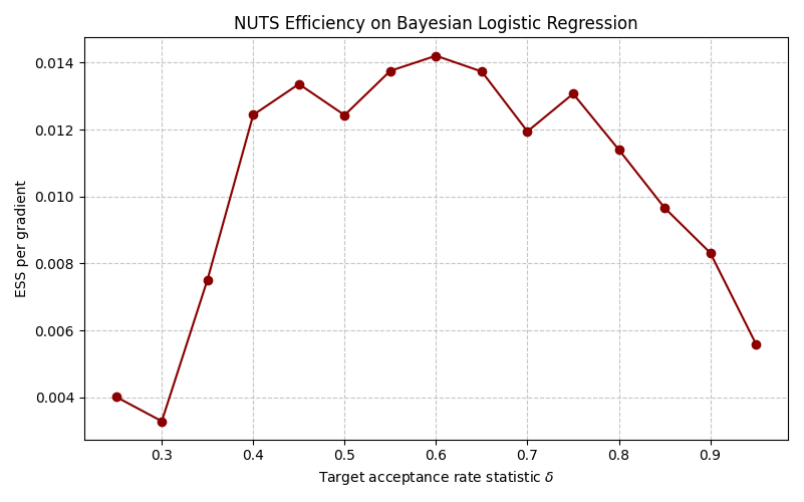


Figure 5: Bayesian Logistic Regression (25-D).

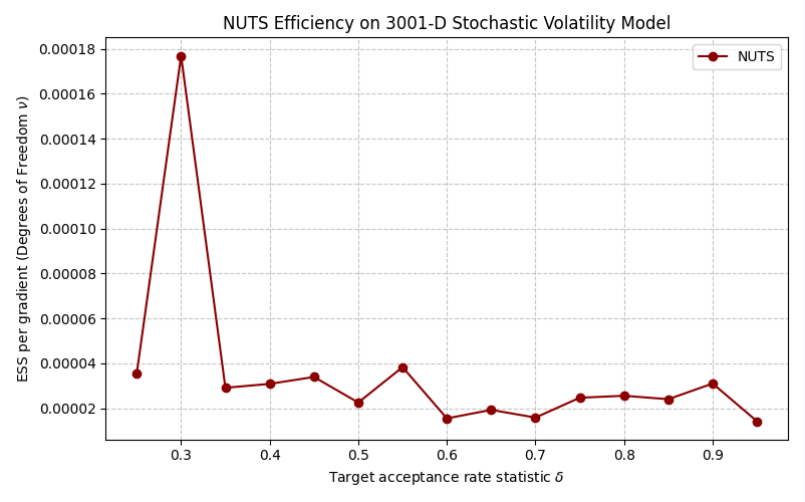


Figure 6: Hierarchical Logistic Regression (302-D).

Appendix: Dual Averaging Convergence

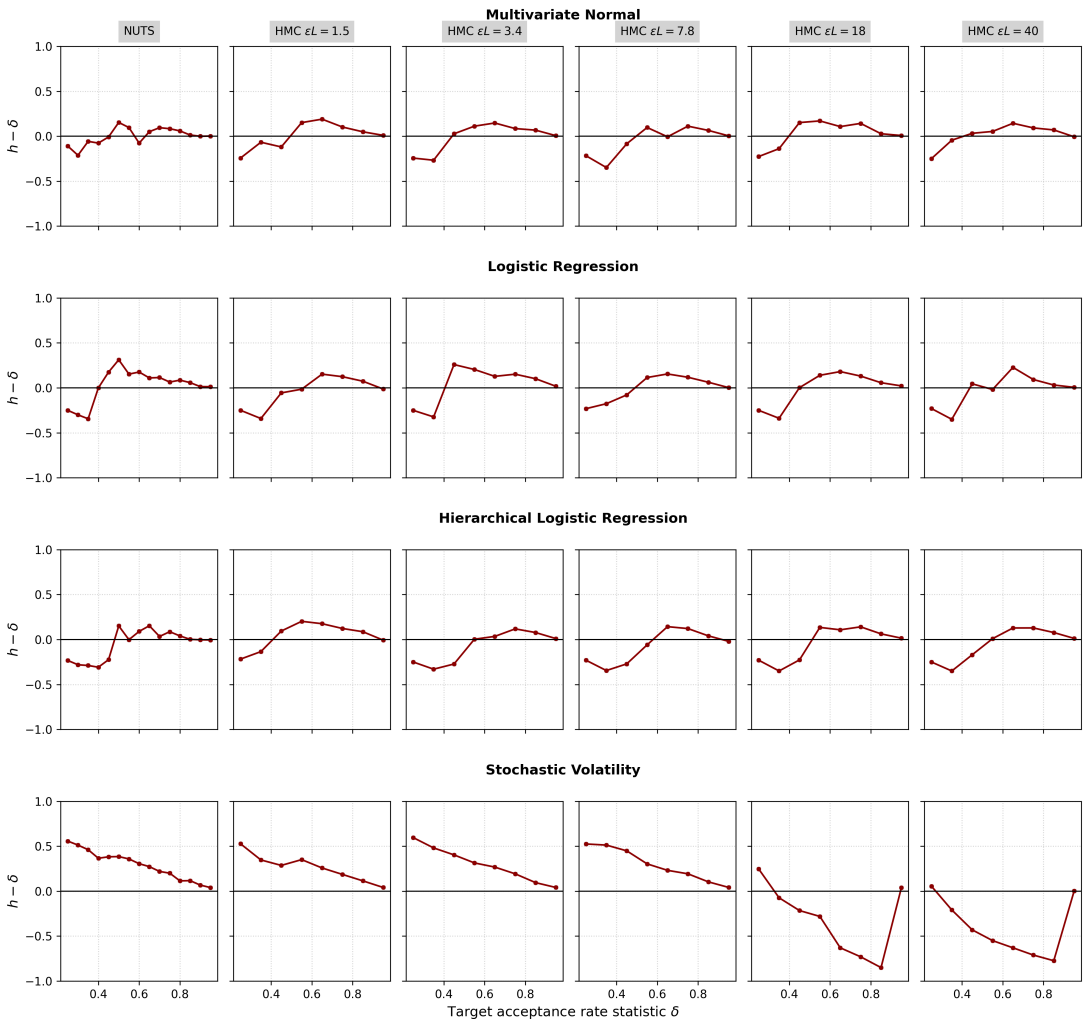


Figure 7: Discrepancy plots ($h - \delta$ across δ) for all four original benchmarks. Dual averaging reliably drives acceptance to target δ within a few hundred warm-up iterations.

Appendix: BNN Adaptation Diagnostics

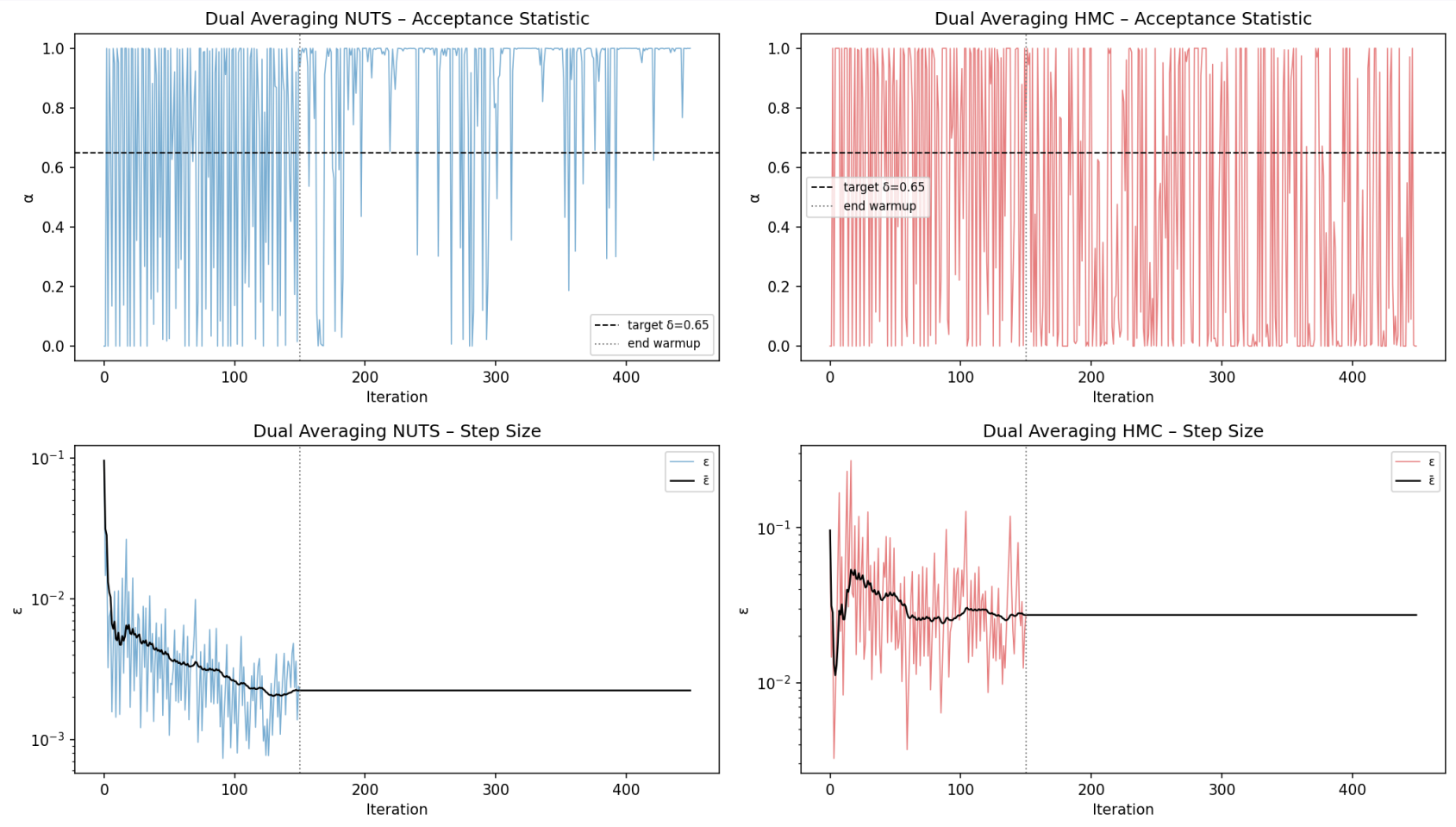


Figure 8: Dual-averaging convergence on the BNN. Step size $\bar{\varepsilon}$ stabilises within the warm-up phase.

BNN: Efficiency

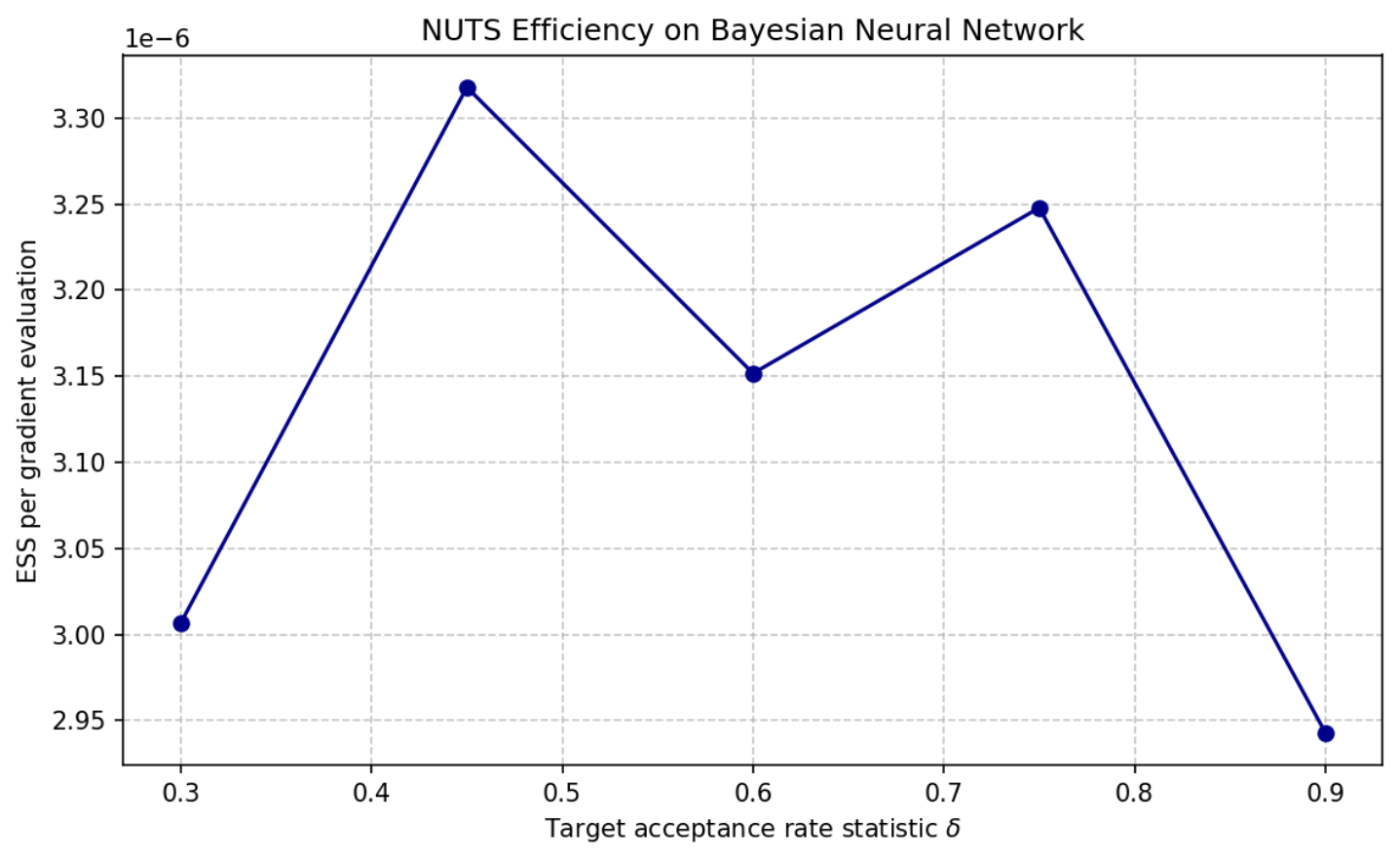


Figure 9: ESS/gradient on the BNN. NUTS consistently achieves higher effective samples per gradient evaluation than HMC across all tested configurations.

Appendix: GPC Kernel Matrix & Posterior Correlations

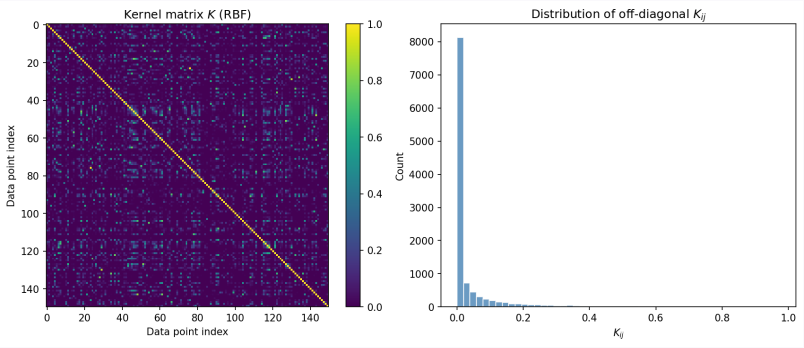


Figure 10: RBF kernel matrix K: dense correlation structure.

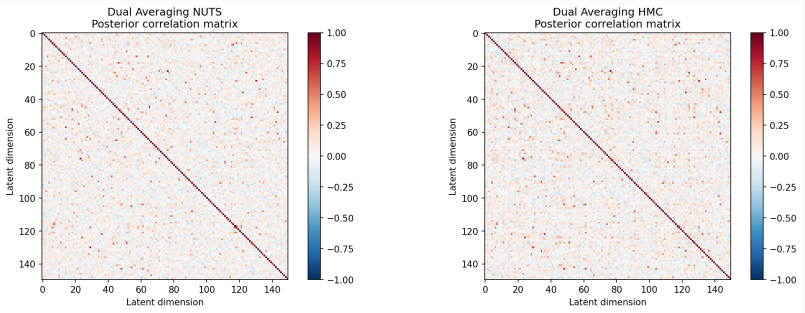


Figure 11: Posterior correlations among latent values f.

Appendix: GPC Efficiency

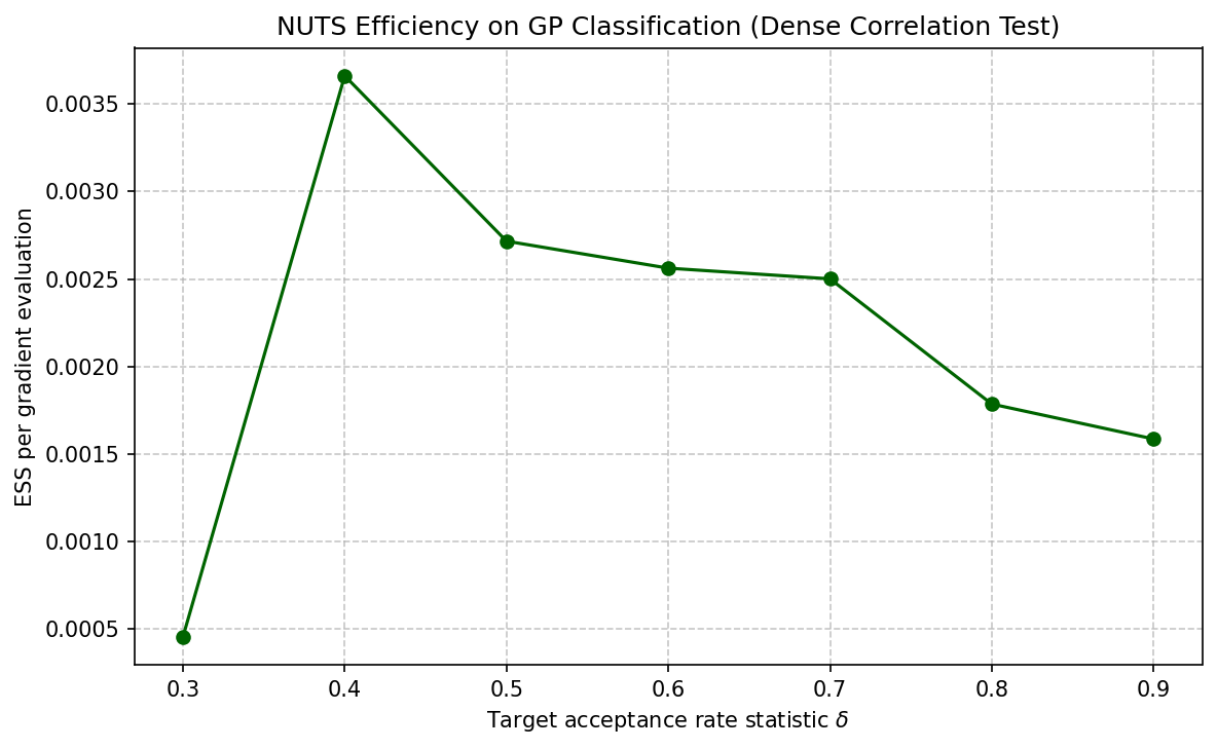


Figure 12: ESS/grad on GP classification. NUTS wins on per-gradient quality but at higher absolute cost.

Appendix: Funnel Trajectory Lengths

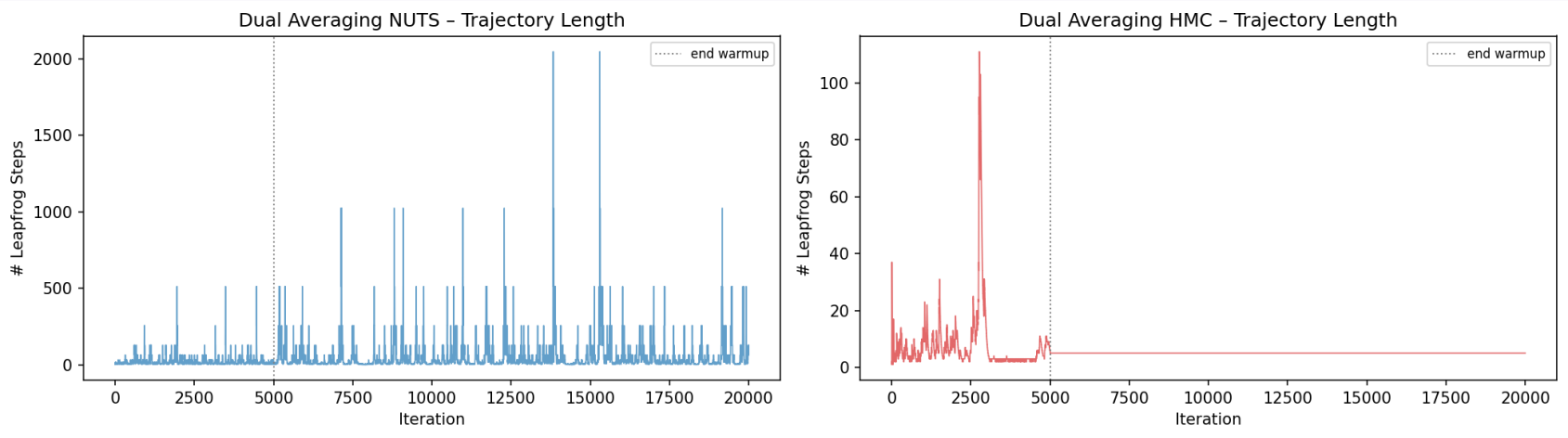


Figure 13: NUTS trajectory lengths on Neal's Funnel. Adaptive mechanism adjusts to local geometry.